

Henry Stephenson

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EDUCATION, AWARDS, & DISTINCTIONS

McGill University – Bioengineering, Minor in Applied AI – 3.98 GPA, Dean's Honor List (every year)	2021-26
• Hatch Scholarship: \$30,000 3-year scholarship awarded by Hatch Ltd. to “outstanding” students in engineering on the basis of academic achievement and community impact on and off campus	2025
• McGill Scarlet Key Award: 1 of 8 undergraduate students inducted in 2025 for demonstration of “excellence, leadership, dedication and perseverance” in taking on a significant project.	2025
• McGill University Prestige Scholar: J W McConnell Award and Major Entrance Scholarship	2021
• Faculty of Engineering Scholarships Fund: granted to academically outstanding students in Engineering; only students academically in top 5% of engineering eligible for the award.	2023
• 2023 NSERC USRA Summer Undergraduate Research in Engineering Award	2023
• McGill Engineering Class of 1953 Scholarship: to an “outstanding undergraduate student”	2022
• Brodeur-Drummond Engineering Scholarship: awarded for “contribution to university life”	2023
• Camillia Samson Engineering Scholarship: awarded on the basis of “high academic standing”	2023
• 2022 BRIDGE Sustainability Case Competition Innovation Award	2022

I E Weldon Secondary School – International Baccalaureate Full Diploma Program	2017-21
• IB Final Score: 44/45 (99 th percentile); highest GPA in class of >300 (98.3% final avg.)	2021
• Lieutenant Governor Volunteer Award (provincially designated); IB CAS Award	2021
• Principal's List Award (top ten average in school of >1200 students)	2018, '19, '20, '21

Select Awards & Distinctions

• Valedictorian for the graduating class of 2021; Valedictorian's Award	2021
• SAT: 1540 (780 Math, 760 Reading & Writing), > 99 th percentile	2020
• 1 of 5 provincial recipients of \$8,000 Ontario Minor Hockey & DFO Scholarship Award	2021
• <u>Accredited Canadian Investment Advisor</u> (completed the Canadian Securities Course (CSC))	2023
• U Waterloo Fermat Math Contest Certificate of Distinction; Math Contest School Champion	2019

WORK EXPERIENCE

Capstone AI Intern – Congruence Therapeutics, Montréal	2025-Now
• Benchmarking state-of-the-art open-source biomolecular structure prediction models to compare their relative performances across various prediction tasks/interaction classes.	
• Developing a CLI from scratch to streamline inference and analysis for scientists of all backgrounds.	
• Reporting on utility of open-source models for drug development and protein engineering applications.	

Summer Researcher under Nobel Laureate David Baker – Institute for Protein Design, Seattle	2025-Now
• Leveraging state-of-the-art computational tools (RFDiffusion2 & 3, Protein/LigandMPNN, AlphaFold3, filtering techniques) and biochemical, physicochemical, and biomimetic insights to engineer de novo proteins for proteolysis of A β 42, tau protein, and other aggregating proteins involved in neurodegenerative diseases like Alzheimer's, Parkinson's, and amyotrophic sclerosis (ALS).	
• Skills: Python (NumPy, Pandas, PyTorch, Matplotlib, Jupyter, Rosetta suite), HPC (Slurm), Bash, Git.	
• Working among world-class researchers to present findings to improve medical outcomes for one of the leading causes of disability and death globally: neurodegeneration.	

Venture Capital Investment Associate – Front Row Ventures (FRV), Montréal	2025
• FRV is a student-focused, student-led, industry-agnostic VC which has funded >60 startups including Valence Discovery, Nomic Bio, and Puzzle Medical.	
• From deal sourcing to due diligence to IC to capital deployment, I supported FRV in funding world-changing student technologies while simultaneously maximizing investor returns.	

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Assay Engineering and R&D Intern – Nomic Bio, Montréal 2023-24

- Youngest person ever offered a position at the startup. Optimizing the nELISA proteomic platform by: developing novel nanotech to maximize Ab fidelity and minimize Ab cross-reactivity; building workflows from scratch to analyze data; coordinating across company divisions to engineer solutions
- Techniques: flow cytometry, immunoassays, DNA modification & nanotechnology, surface chemistry

Research Assistant – Noémie Dorval Lab, McGill University 2023-Now

- 2nd author on “Engineered Heat-Inducible Living Textiles”, published in ACS Synthetic Biology
- Developing modular bio-based heat sensor in *E. coli* for diverse applications: body temperature monitoring, heat-based tunable cancer therapy, bio-encryption, bioreactor thermal modulation, etc.

HIV Project Junior Lead, Research Assistant – Mario Ostrowski Lab, University of Toronto 2022

- Major contributor to “Immunogenicity of COVID-19 vaccines and their effect on the HIV reservoir in older people with HIV” paper; published in Cell Press – iScience
- Elucidated SARS-CoV-2 vaccine safety for over 39 million people globally with HIV, providing scientific assurance that vaccination is not harmful; trained & led team of 5 undergrads in experiments

Research Assistant – Natalie Reznikov Lab, McGill University 2023

- SURE project under Prof. Natalie Reznikov: “Synbio approaches for construction of Biobricks”
- McGill SURE Award for Best Research Presentation with over 100 participating SURE researchers
- Developing therapeutic to save patients with pancreatic cancer. Re-engineering novel CRISPR-Cas system to induce pyroptosis (inflammatory cell death) in cancer cells, leaving healthy cells unharmed

EXTRACURRICULAR & LEADERSHIP EXPERIENCE

iGEM, 2023 Grand Prize Winners, >400 Teams – Team Lead, Finance Director, Advisor 2022-25

- McGill iGEM: First Canadian team to win 1st place worldwide for our CRISPR-Cas pancreatic cancer therapeutic *PROTEUS* (provisional patent under review); presented to >5000 attendees
- Jointly managed team of 30; developed students' wet lab skills, from pipetting up to western blots
- As the Finance Director, I wrote our Business Plan, a >100-page report on our commercialization strategy post-patenting; also procured > \$150,000 in funding with over 50 grant submissions

In Vitro Biotech Conference, Co-Lead Coordinator 2022-23

- Created a first-ever Canada-wide biotechnology conference hosted in Montreal with over 250 registered participants and over \$20,000 in expendable funding.
- Procured funding, garnered interest from all facets of the bioeconomy from VCs to big pharma, and coordinated catering and venue booking to connect Canadian biotech on a national level.
- Speaker and moderator on “The Role of VCs in Facilitating Change in Biotech” panel.

McGill Classical Music Club, President, previously VP Finance 2022-23

- Manage McGill's preeminent organization for top-performing classical musicians; arrange chamber groups, procure funding, organize concerts & performances in old-person homes

McGill Biomechanics Club, Mechanical Engineering Team 2021-22

- Played integral role in mechanical design strategy for development of novel magneto-rheological tremor suppressor for Parkinson's disease, using Python and CAD in ideation and design

SKILLS & INTERESTS

Languages: Fluent English, intermediate (B1-B2) French, intermediate German and Chinese (Mandarin).

Skills: accredited investment advisor, programming (Python, Java, Clojure, C, Bash, SLURM, Git(Hub)), data science / ML (NumPy, Pandas, PyTorch, Matplotlib/Seaborn, etc.), bioinformatics, lab research.

Interests: Protein engineering and computational biology, synthetic & molecular biology, machine learning, linguistics, language-learning, piano, soccer, hockey, cross-country.

Events Attended: SynBioBeta 2023 (helped organize as a volunteer lead), SynBio Canada 5.0 through 7.0.